Antibiotic Man

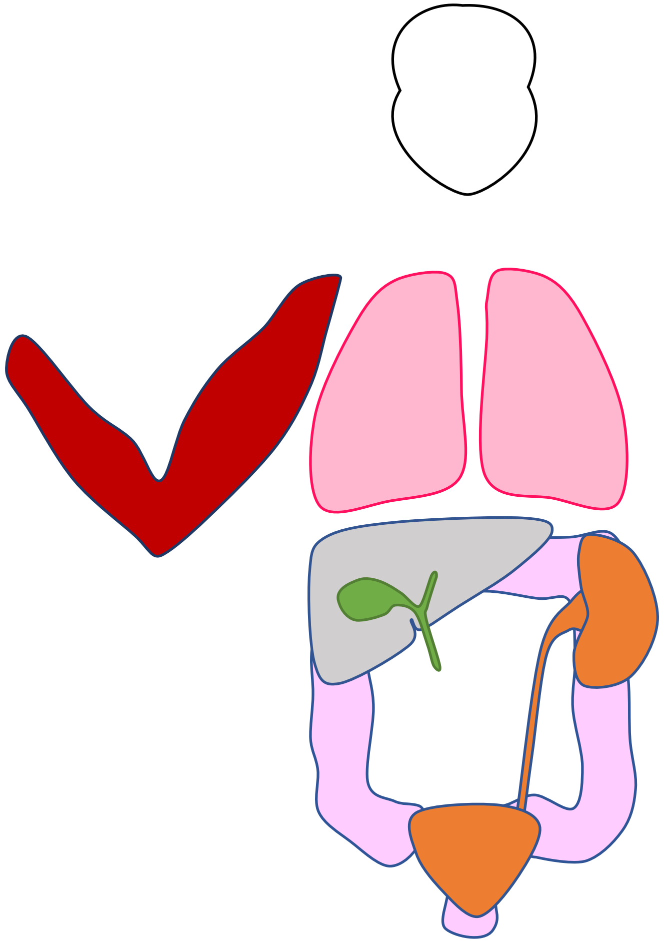
Pneumonia

UTI

Cellulitis

Intraabdominal

Cases



Antibiotics - Part 2: Sites of Infection

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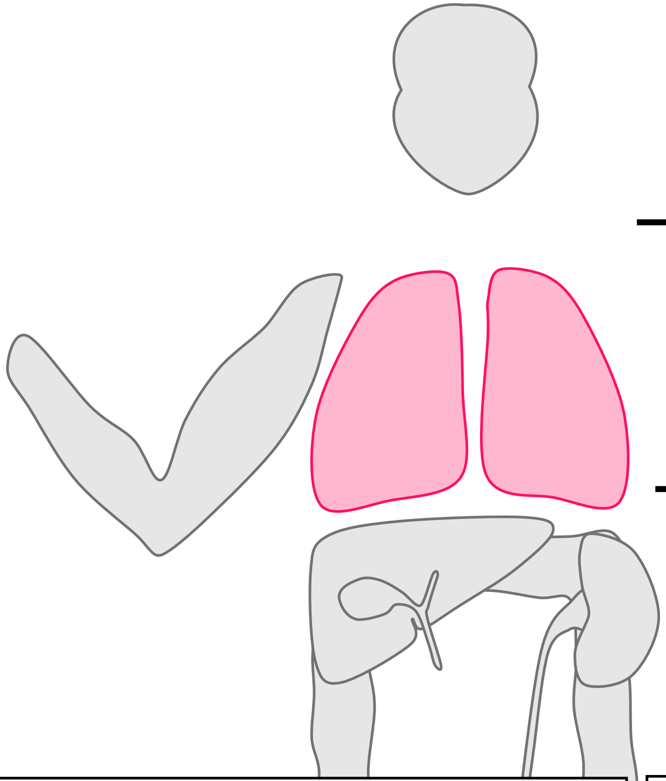
PNEUMONIA



	Bugs	Drugs
CAP	CAP Bugs *S. pneumoniae* GNRs (*H. flu, Moraxella*) Atypicals	CAP Drugs Ceftriaxone + azithromycin/doxycycline Respiratory FQ
HAP/ VAP (>48h)	HAP/VAP Bugs + *P. aeruginosa* + MDR GNRs ± MRSA	HAP/VAP Drugs Zosyn/ cefepime/ meropenem/ aztreonam/ levofloxacin ±Vancomycin/ linezolid
Aspiration	Aspiration Bugs *S. pneumoniae*/ GPCs Enteric GNRs Oral anaerobes	Aspiration Drugs CAP ± anaerobic coverage *If covering anaerobes:* Unasyn/ clindamycin/ moxifloxacin

The **2019 IDSA/ATS CAP guidelines** do not recommend routinely covering for anaerobes for aspiration pneumonia unless:

1) Lung abscess
2) Empyema

The **2019 IDSA/ATS CAP guidelines** abandoned HCAP as a separate

Empirically cover for MRSA and Pseudomonas in patients with CAP + risk factors. See HAP/VAP drugs for empiric therapies.

1) Prior colonization with MRSA or P. aeruginosa
2) Hospitalization with IV abx within 90 d

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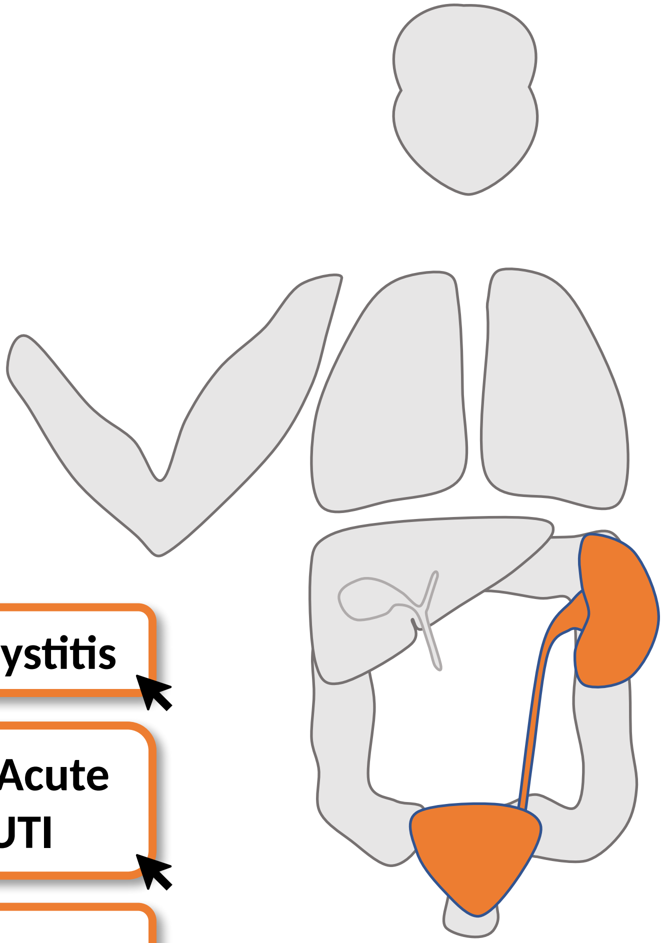
Any patient with a UTI may be at risk for MDR organisms. Risk factors include (< 3 mos):

- Prior infection with MDR/ESBL organism
- Inpatient stay or care facility
- Travel to area with high rates of MDR organisms
- Use of FQ, Bactrim or broad-spectrum β-lactam



Uncomplicated cystitis

Pyelonephritis/ Acute complicated UTI

CAUTI

Risk for MDR?

Bugs	**Drugs**	
Staph (MSSA/MRSA)	Nitrofurantoin	1[st] line for acute cystitis
***E.** coli 75-95%*	Bactrim	
Enterococcus	Fosfomycin	
Klebsiella	Urinary FQ	
	β-lactam (e.g., Augmentin)	
Proteus		
Pseudomonas	Ceftriaxone	
	Zosyn	
	Carbapenems	
	May need vancomycin, daptomycin or linezolid for *Enterococcus*	

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CAUTI = symptomatic bacteriuria with a suprapubic, indwelling, intermittent cath within the past 48 h.

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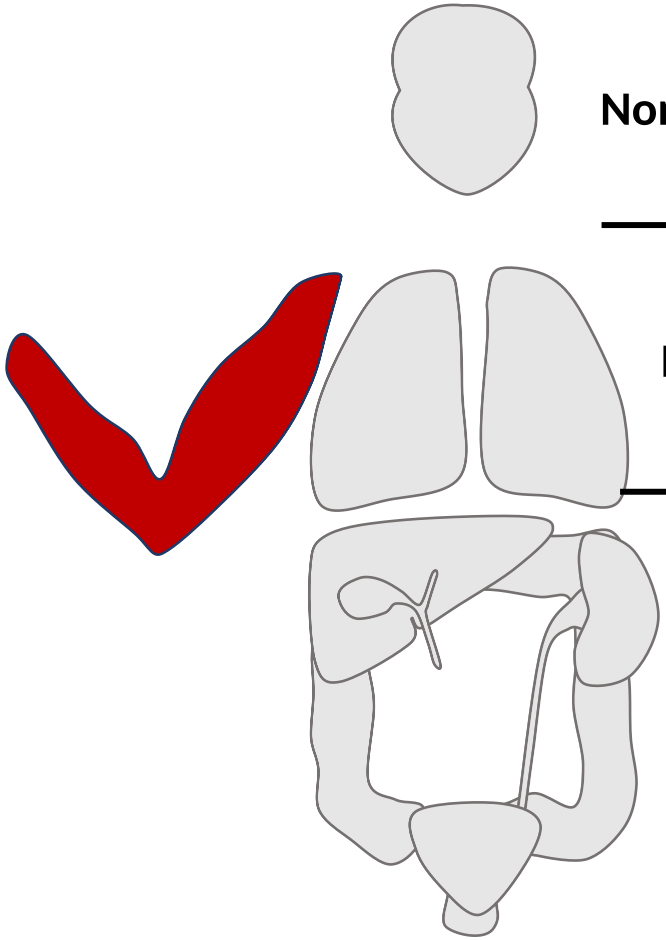
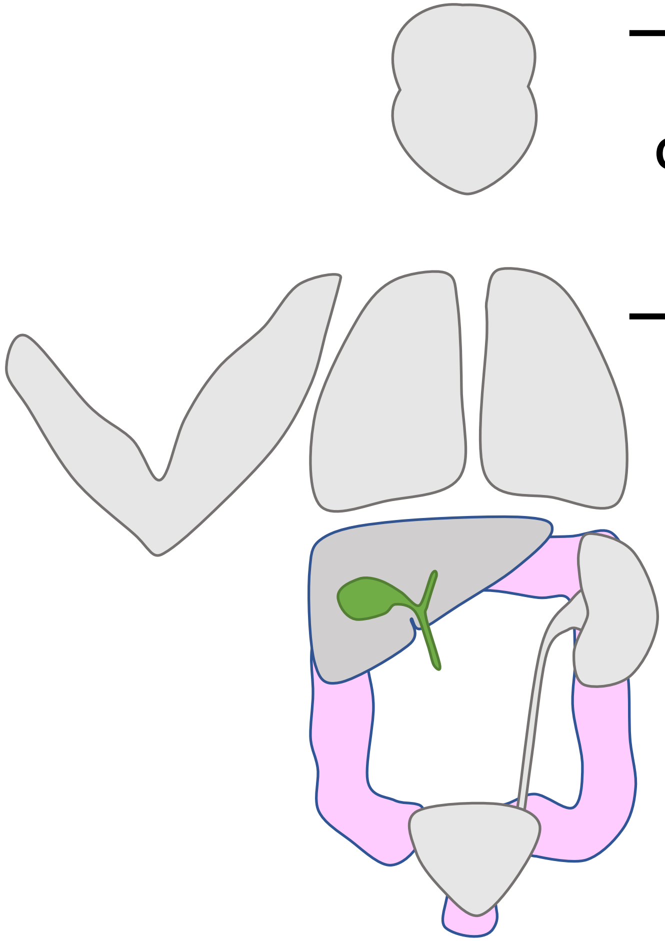
SKIN AND SOFT TISSUE INFECTIONS



	Bugs	Drugs
Non-purulent	GPCs Nonpurulent Bugs (*Strep* >> *S. aureus*)	Cefazolin/ cephalexin Nonpurulent Drugs
Purulent	MSSA/ MRSA Purulent Bugs GAS	Vancomycin/ daptomycin Purulent Drugs Bactrim/ doxycycline ± cephalexin
NSTI	GAS > *S. aureus* GNRs (*Aeromonas, Vibrio*) NSTI Bugs Anaerobes (*Clostridium*) Polymicrobial	Vancomycin /daptomycin NSTI Drugs + Zosyn/ meropenem + clindamycin

Treatment for NSTI includes early surgical debridement. Surgery should be consulted on all patients with suspected NSTI. The addition of clindamycin provides antitoxin effect.

INTRAABDOMINAL INFECTIONS



	Bugs	Drugs
Community acquired	Enteric GNRs *Streptococci* CA-IAI Bugs Anaerobes (*B. fragilis*)	3° cephalosporin + metronidazole CA-IAI Drugs FQ + metronidazole
Hospital acquired	+ *P. aeruginosa* + Enterococcus Purulent Bugs + CoNS, S. aureus	Zosyn/ cefepime + metronidazole/ meropenem Purulent Drugs ± Vancomycin

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1-3

4-5

Case 1

70 yo presents to ED with 4 days of productive cough, malaise and fevers. T 100.9F, HR 110, BP 99/53, RR 33, SpO2 90%.

WBC is 17k, lactate is 2.1. CXR is notable for a focal consolidation in the RLL.

CAP. Based on his vital signs and age (CURB-65), he should be managed in the hospital.

S. pneumo is most common pathogen. **GNRs** (*H. flu*, *Moraxella*), **atypicals**.

Ceftriaxone and azithromycin. Or a **respiratory FQ** (levofloxacin, moxifloxacin.

Ceftriaxone + doxycycline. Doxycycline can be used for atypical coverage when there are QT concerns.

Ceftriaxone/ azithromycin ± Vancomycin (if severe pneumonia). At risk for MDR given 1) history of MRSA or PsA or 2) hospitalization + IV abx in 90 d.

Case 2

30 yo with IVDU presents with a painful swollen arm after injecting into his arm 2 days ago. T is 99F, HR 100s, BP 100/60.

Their L arm is red, warm, and tender. There is a 2x2 cm area of fluctuance.

Purulent SSTI

***S. aureus* (MRSA > MSSA).**

Antibiotics + I&D. **Bactrim, doxycycline** is generally > clindamycin Additional Strep coverage with cephalexin can be considered in patients at high risk of endocarditis.

...Less than 24 h later, they present to the ED with fevers/ chills. They are febrile to 102F, HRs 120s, BP 100/60.

IV vancomycin.

Case 3

21 yo presents with abdominal pain, nausea/vomiting. VS show T of 103F, HR 140s, BP 95/55, SaO2 99% on RA. They have a rigid abdomen and have rebound on exam. CT shows signs of appendicitis.

Enteric GNRs, **streptococci**, enteric **anaerobes** (*Bacteroides*).

Ceftriaxone + metronidazole

...They are managed with an appendectomy and antibiotics. Two weeks later, they represent with fevers, abdominal pain and are found to have an intraabdominal abscess.

Pseudomonas, ***Enterococcus*** and ***S. aureus*** are more common in **hospital acquired/ healthcare associated IAI**.

Zosyn ± vancomycin. Based on recent hospitalization and surgery, she is considered to have MRSA risk factors.

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Case 4a

65 yo with a history of stroke with residual L hemiparesis and residual dysphagia presents from SNF with confusion.

VS: T 100.5F, HR 110s, BP 110/60s. Grimaces with palpation of suprapubic region and L flank.

WBC 17k, lactate 2.1. CXR is clear. UA shows + LE, 50 WBCs, many bacteria.

Suspected pyelonephritis/ complicated UTI.

E. coli is responsible for 75-95% of UTIs. Enteric GNRs and some *Staph* spp. are also common.

Ceftriaxone. While he does have risk factors for MDR organisms (SNF) it is reasonable to start narrower here.

This is a **CAUTI**. Additional bugs include **PsA** and ***Enterococcus***. IV ceftriaxone or IV Zosyn (for broader coverage).

IV ertapenem.

Case 4b

65 yo with a history of stroke is readmitted 2 months later with cough and shortness of breath and is admitted to the ICU. CXR shows a RLL consolidation.

CAP vs. aspiration pneumonia

Hospitalization + IV abx w/in 90 d is a risk factor for **MRSA** or **PsA**. CAP bugs (***S. pneumo***, **GNR, atypicals**) and **anaerobes** are also possible.

Vancomycin + cefepime + azithromycin. Patients with severe pneumonia should receive empiric coverage of MRSA + PsA. Routine coverage of anaerobes is not necessary.

Add **metronidazole** or switch to **Vancomycin + Zosyn + azithromycin**. Cover anaerobes with lung abscess or empyema.

Case 5

33 yo presents with rapidly ascending erythema in the L leg that started 12 h prior. T 103F, HR 150s, BP 80s/40s. Exam is notable for deep erythema extending from ankle to mid thigh with areas of duskiness. They have tenderness beyond the area of erythema, no palpable fluctuance.

Na is 123, Cr 3.5 (b/l is 1.0), WBC 25k, lactate 6.7.

Necrotizing soft tissue infection

GAS, *S. aureus*, GNRs, anaerobes (*Clostridium*). May be polymicrobial.

Vancomycin + Zosyn/meropenem + clindamycin. Surgery should be consulted.

2023-2025 Practice Updates - Sites of Infection

2025 ATS CAP guideline (AnnalsATS / AJRCCM 2025):

- Selective empiric MRSA/Pseudomonas coverage - cover only with validated risk factors or a locally validated risk score; de-escalate at 48 h if cultures/screens negative.
- Shorter courses - 3-5 days for inpatients clinically stable by 48-72 h (afebrile, <=1 instability sign).
- Corticosteroids for SEVERE CAP - hydrocortisone <400 mg IV/day x5-7 d (CAPE COD, NEJM 2023; mortality RR 0.62). NOT for non-severe CAP; use dexamethasone if COVID-19.
- MRSA nares PCR - high negative predictive value; use it to stop empiric vancomycin.

UTI - 2024 WikiGuidelines (JAMA Netw Open 2024) & EAU 2024-25:

- Cystitis/UTI definitions no longer sex-dependent - uncomplicated cystitis can be diagnosed in men.
- First-line cystitis unchanged: nitrofurantoin x5 d, TMP-SMX x3 d (if resistance <20%), fosfomycin x1.

SSTI / intra-abdominal - IDSA 2014 SSTI & SIS 2017 IAI frameworks on this board remain current; first-line regimens unchanged.

Take-Home Points

Healthcare-associated infections add MRSA, Pseudomonas, and Enterococcus - escalate empiric coverage only with risk factors.

CAP: S. pneumoniae + GNRs + atypicals -> ceftriaxone + azithromycin (or respiratory FQ). HAP/VAP (>48 h) adds MRSA + Pseudomonas.

Aspiration and former-HCAP patients: treat as routine CAP; add anaerobic coverage only for lung abscess or empyema.

UTI: E. coli predominates. CAUTI and complicated UTI add Pseudomonas, Enterococcus, Staph. Nitrofurantoin/fosfomycin only for cystitis, never pyelonephritis.

SSTI: nonpurulent = Strep (cefazolin/cephalexin); purulent = S. aureus/MRSA (I&D + vancomycin or PO MRSA agent); NSTI = surgery + broad coverage + clindamycin.

Intra-abdominal: community = enteric GNRs/anaerobes (ceftriaxone+metronidazole); hospital-acquired adds Pseudomonas/Enterococcus. Always pair antibiotics with source control.

Attribution & Changelog

Adapted from TeachIM "Antibiotics Part 2: Sites of Infection" (teachim.org), published Sept 2021.

Authors: Yilin Zhang MD; Brandon Fainstad MD; Mike Hori MD.

Source: teachim.org/teaching_material/abx-2-sites/ | Deck: Antibiotics-Sites-of-Infection-4.22-1.pptx

Corrections (this optimization, 2026-06-10):

Slide 2 (Pneumonia): "doxyclcine" -> "doxycycline" (spelling).

Slide 8 (summary table): "S. aereus" -> "S. aureus" (spelling).

Added: 2023-2025 practice-update slide, take-home slide, this attribution slide. All original click-to-reveal animations and graphics preserved.

Companion: Common-sites-of-infection-Fill-in-worksheet.docx (learner "Antibiotic man" handout) - reviewed, no changes.

Newer context is source-cited; medical content should be clinician-reviewed before teaching.